

FES ARAGÓN UNAM

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INGENIERÍA EN COMPUTACIÓN

SEGUNDO SEMESTRE

PROGRAMACIÓN ORIENTADA A OBJETOS

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GRUPO 2260

FECHA: 03/05/26

EJERCICIOS REALIZADOS:

TAREA 13 – PRUEBA 1

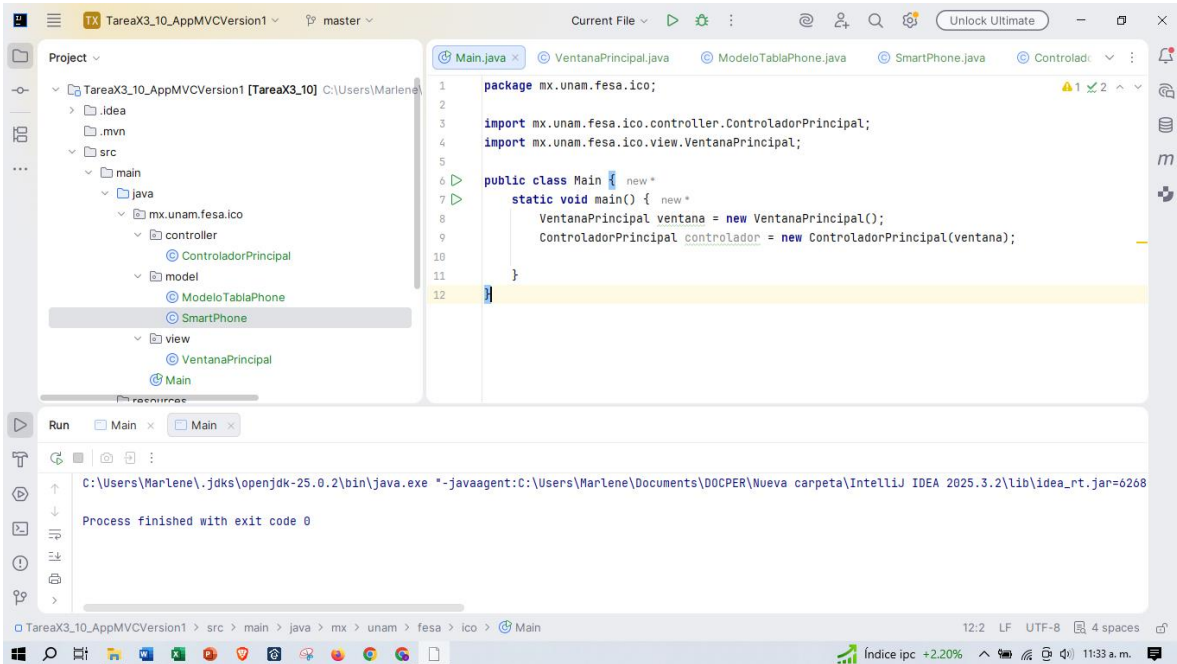
TAREA 13 – PRUEBA 2

TAREA 13 – PRUEBA 3

TAREA 13 – PRUEBA 4

TAREA 13 – PRUEBA 5

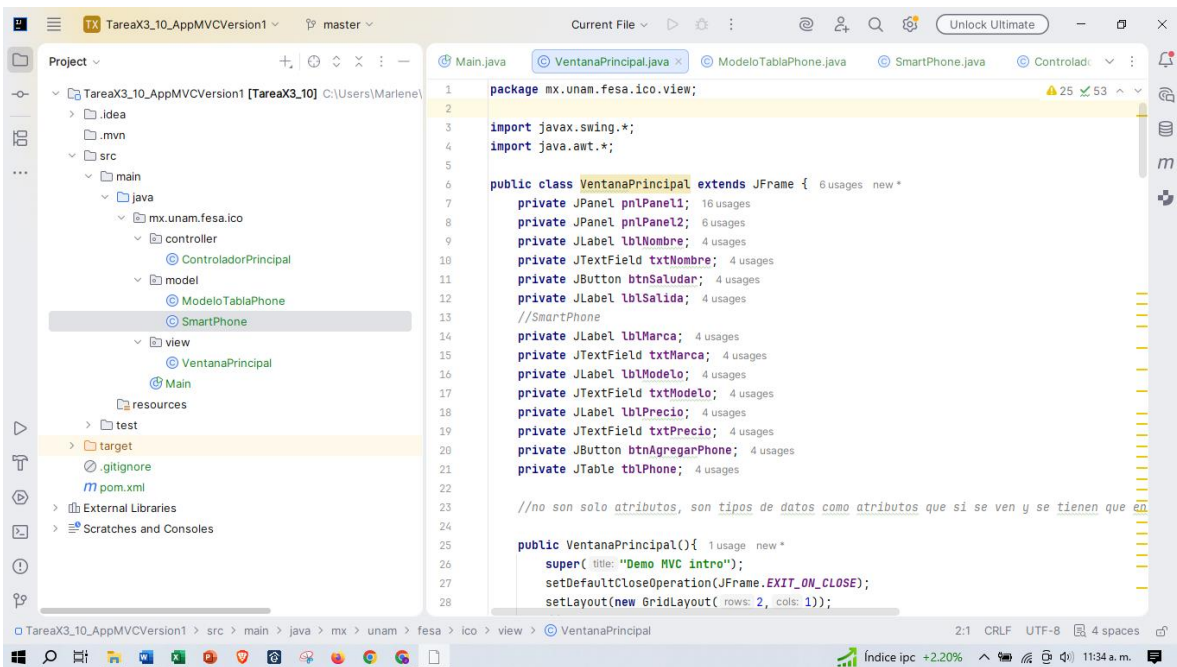
TAREA 13 – PRUEBA 1, 2, 3, 4, 5



The screenshot shows the IntelliJ IDEA IDE with the project 'TareaX3_10_AppMVCVersion1' open. The 'Project' view on the left shows the package structure: `mx.unam.fesa.ico` (containing `controller`, `model`, and `view`), `SmartPhone`, and `Main`. The `Main.java` file is open in the editor, showing the following code:

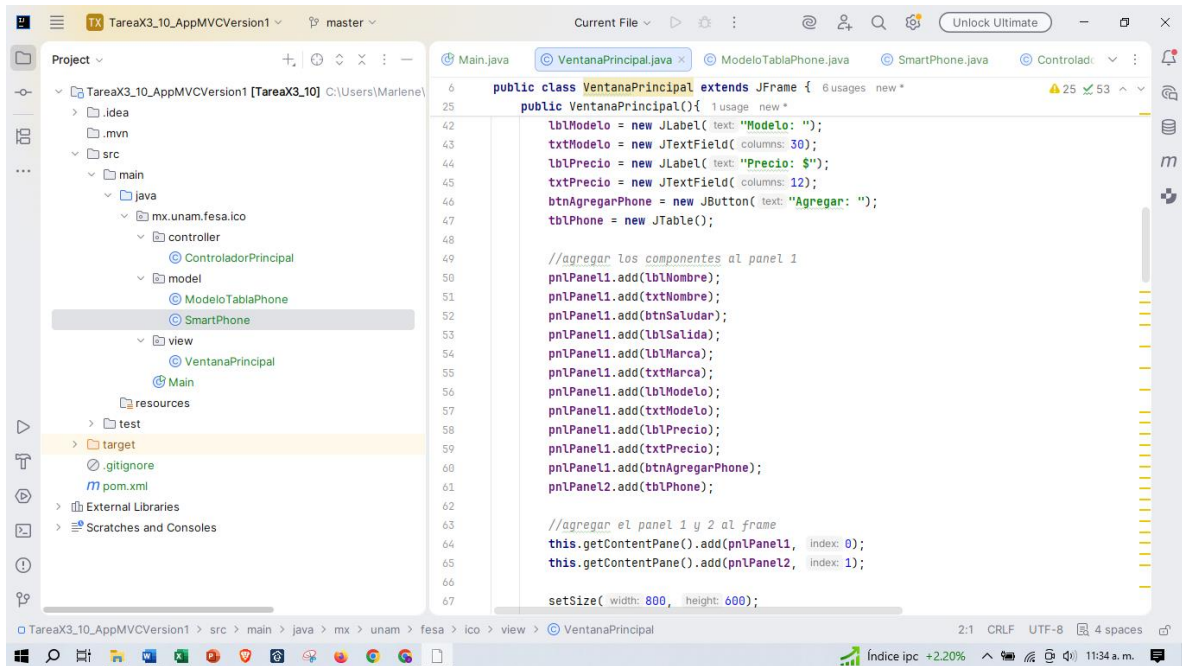
```
1 package mx.unam.fesa.ico;
2
3 import mx.unam.fesa.ico.controller.ControladorPrincipal;
4 import mx.unam.fesa.ico.view.VentanaPrincipal;
5
6 public class Main {
7     static void main() {
8         VentanaPrincipal ventana = new VentanaPrincipal();
9         ControladorPrincipal controlador = new ControladorPrincipal(ventana);
10    }
11 }
12
```

The 'Run' view at the bottom shows the command executed: `C:\Users\Marlene\.jdk\openjdk-25.0.2\bin\java.exe -javaagent:C:\Users\Marlene\Documents\DOCPER\Nueva carpeta\IntelliJ IDEA 2025.3.2\lib\idea_rt.jar=6208`. The process finished with exit code 0.

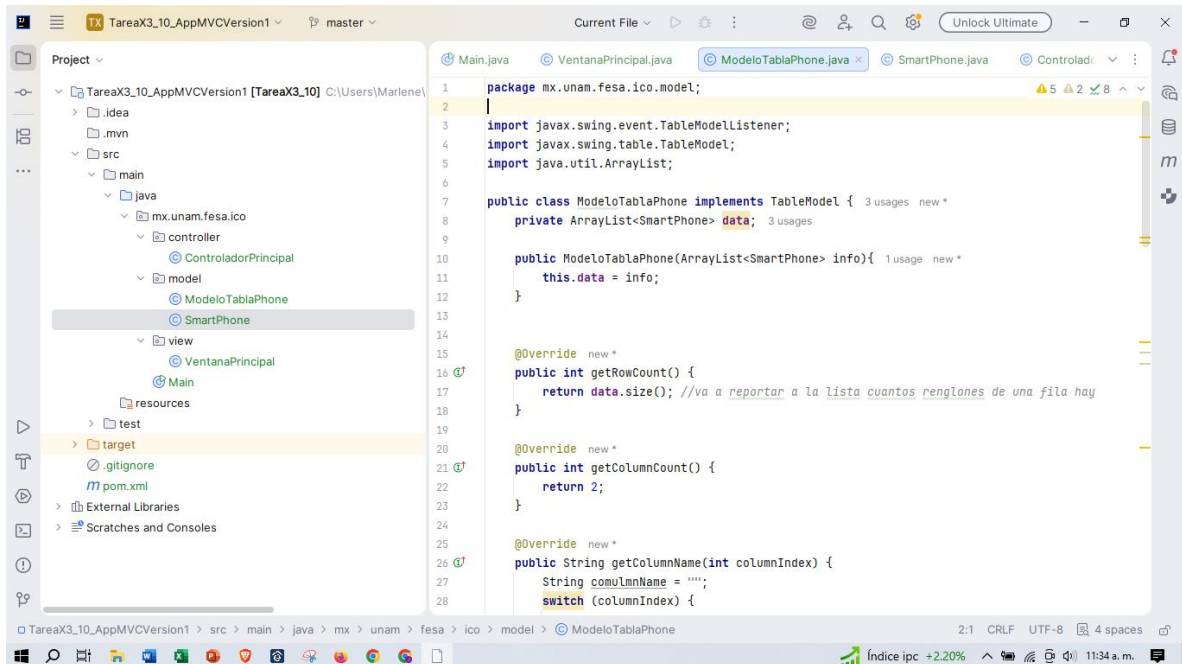


The screenshot shows the IntelliJ IDEA IDE with the project 'TareaX3_10_AppMVCVersion1' open. The 'Project' view on the left shows the package structure: `mx.unam.fesa.ico` (containing `controller`, `model`, and `view`), `SmartPhone`, and `Main`. The `VentanaPrincipal.java` file is open in the editor, showing the following code:

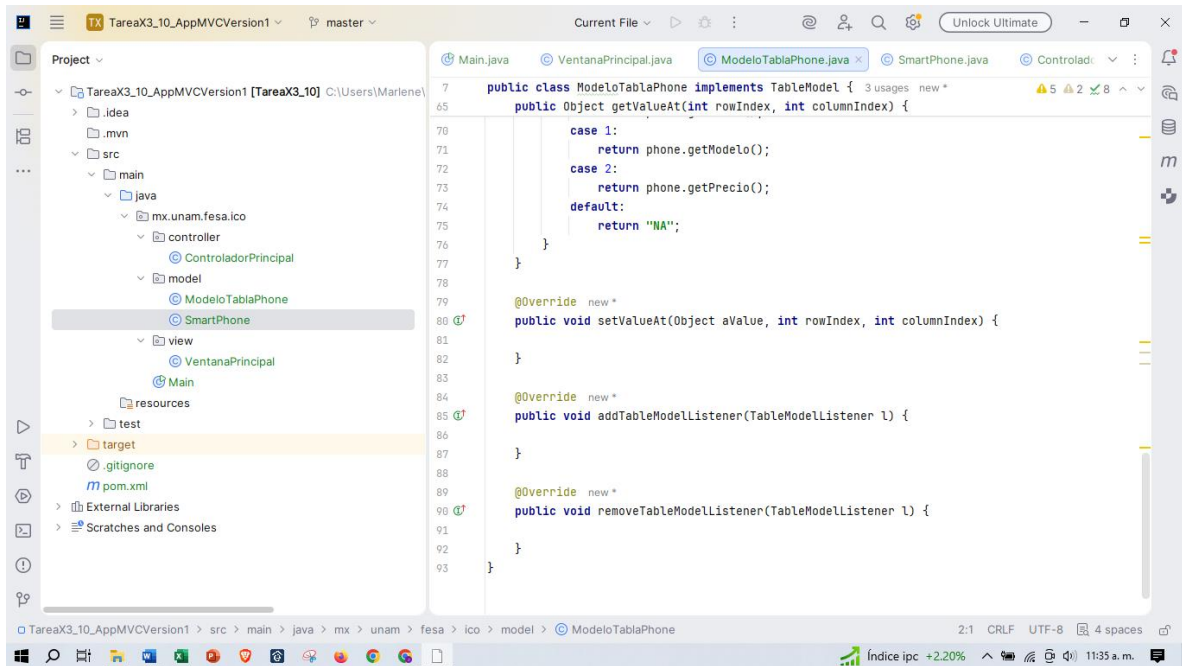
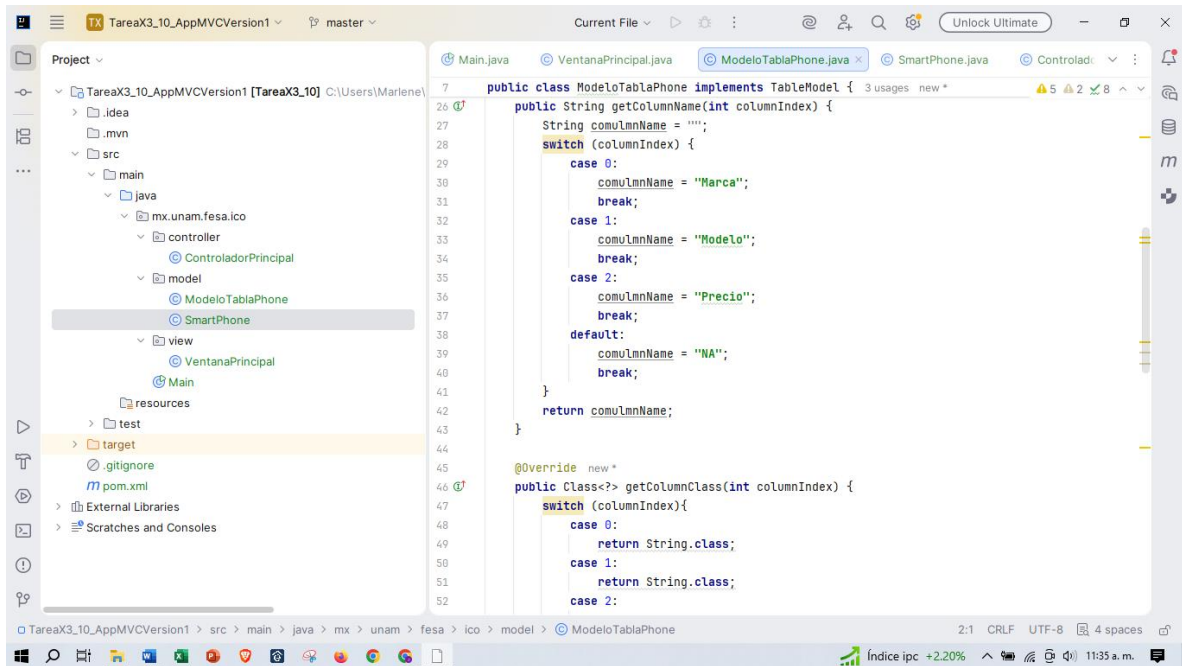
```
1 package mx.unam.fesa.ico.view;
2
3 import javax.swing.*;
4 import java.awt.*;
5
6 public class VentanaPrincipal extends JFrame {
7     private JPanel pnlPanel1;
8     private JPanel pnlPanel2;
9     private JLabel lblNombre;
10    private JTextField txtNombre;
11    private JButton btnSaludar;
12    private JLabel lblSalida;
13    //SmartPhone
14    private JLabel lblMarca;
15    private JTextField txtMarca;
16    private JLabel lblModelo;
17    private JTextField txtModelo;
18    private JLabel lblPrecio;
19    private JTextField txtPrecio;
20    private JButton btnAgregarPhone;
21    private JTable tblPhone;
22
23    //no son solo atributos, son tipos de datos como atributos que si se ven y se tienen que
24
25    public VentanaPrincipal() {
26        super("Demo MVC intro");
27        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
28        setLayout(new GridLayout(2, 1));
29    }
30 }
```

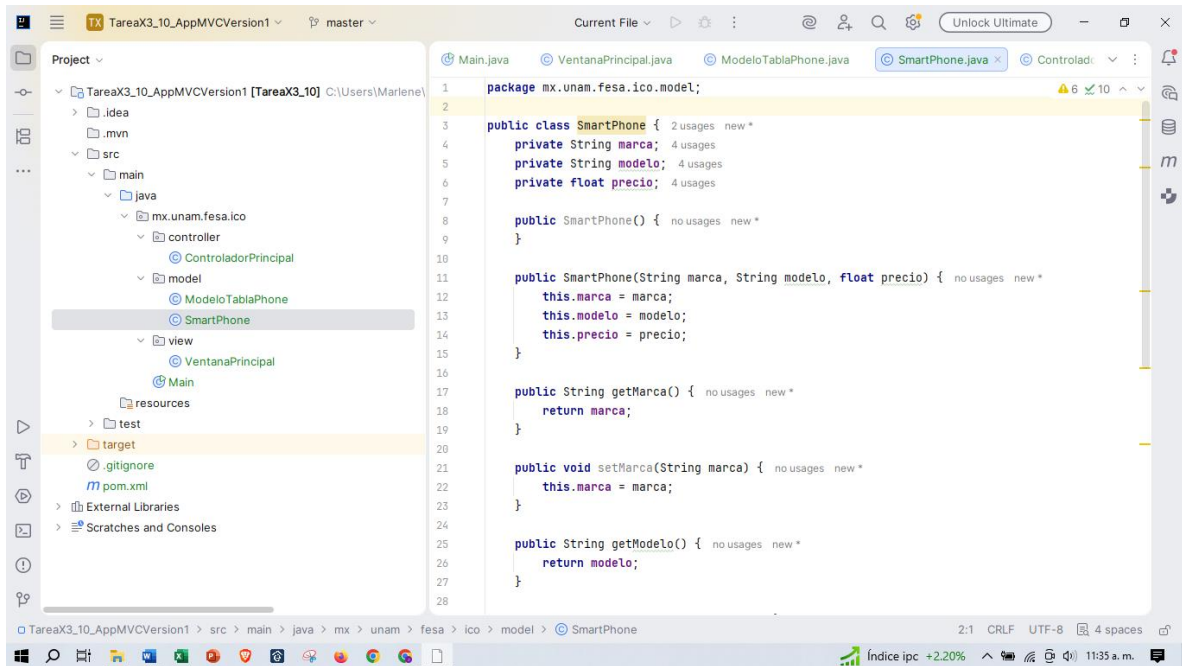


```
6 public class VentanaPrincipal extends JFrame {
7     // ...
8     public VentanaPrincipal() {
9         lblModelo = new JLabel("Modelo: ");
10        txtModelo = new JTextField(30);
11        lblPrecio = new JLabel("Precio: $");
12        txtPrecio = new JTextField(12);
13        btnAgregarPhone = new JButton("Agregar: ");
14        tblPhone = new JTable();
15
16        //agregar los componentes al panel 1
17        pnlPanel1.add(lblNombre);
18        pnlPanel1.add(txtNombre);
19        pnlPanel1.add(btnSaludar);
20        pnlPanel1.add(lblSalida);
21        pnlPanel1.add(lblMarca);
22        pnlPanel1.add(txtMarca);
23        pnlPanel1.add(lblModelo);
24        pnlPanel1.add(txtModelo);
25        pnlPanel1.add(lblPrecio);
26        pnlPanel1.add(txtPrecio);
27        pnlPanel1.add(btnAgregarPhone);
28        pnlPanel2.add(tblPhone);
29
30        //agregar el panel 1 y 2 al frame
31        this.getContentPane().add(pnlPanel1, index: 0);
32        this.getContentPane().add(pnlPanel2, index: 1);
33
34        setSize( width: 800, height: 600);
35    }
36}
```



```
1 package mx.unam.fesa.ico.model;
2
3 import javax.swing.event.TableModelListener;
4 import javax.swing.table.TableModel;
5 import java.util.ArrayList;
6
7 public class ModeloTablaPhone implements TableModel {
8     private ArrayList<SmartPhone> data;
9
10    public ModeloTablaPhone(ArrayList<SmartPhone> info) {
11        this.data = info;
12    }
13
14    @Override
15    public int getRowCount() {
16        return data.size(); //va a reportar a la lista cuantos renglones de una fila hay
17    }
18
19    @Override
20    public int getColumnCount() {
21        return 2;
22    }
23
24    @Override
25    public String getColumnNames(int columnIndex) {
26        String comuLmnName = "";
27        switch (columnIndex) {
28            // ...
29        }
30    }
31}
```

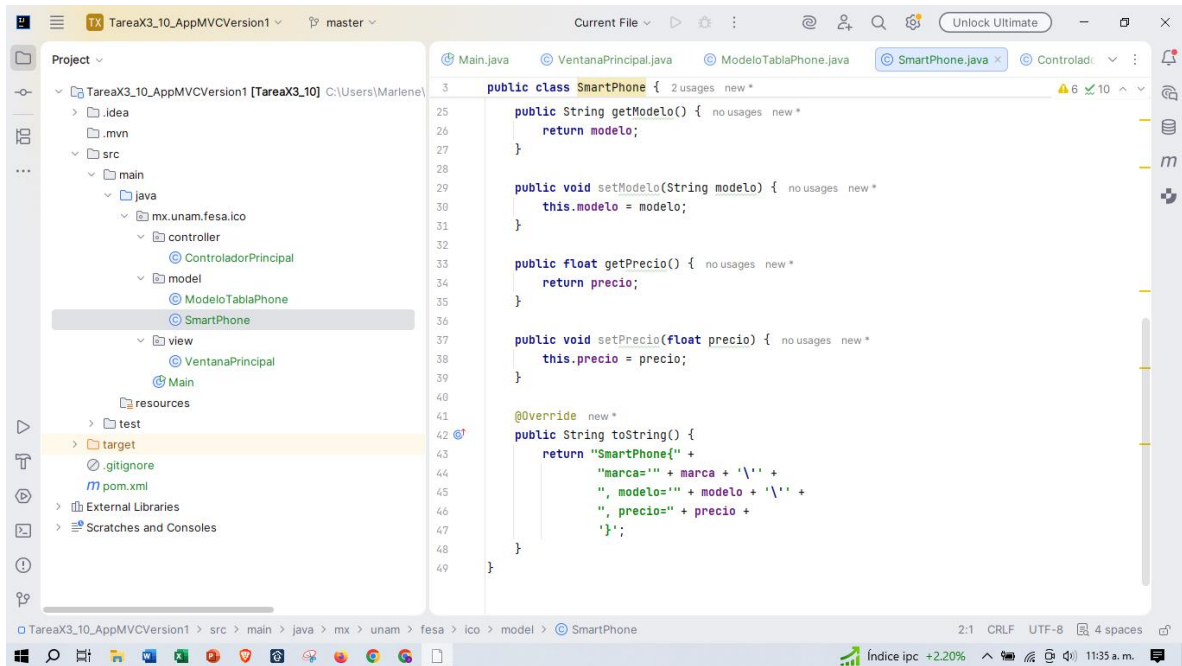




Project: TareaX3_10_AppMVCVersion1 [TareaX3_10] C:\Users\Mariel\...

Package: mx.unam.fesa.ico.model;

```
1 package mx.unam.fesa.ico.model;
2
3 public class SmartPhone { 2 usages new *
4     private String marca; 4 usages
5     private String modelo; 4 usages
6     private float precio; 4 usages
7
8     public SmartPhone() { no usages new *
9     }
10
11     public SmartPhone(String marca, String modelo, float precio) { no usages new *
12         this.marca = marca;
13         this.modelo = modelo;
14         this.precio = precio;
15     }
16
17     public String getMarca() { no usages new *
18         return marca;
19     }
20
21     public void setMarca(String marca) { no usages new *
22         this.marca = marca;
23     }
24
25     public String getModelo() { no usages new *
26         return modelo;
27     }
28 }
```



Project: TareaX3_10_AppMVCVersion1 [TareaX3_10] C:\Users\Mariel\...

```
25 public class SmartPhone { 2 usages new *
26     public String getModelo() { no usages new *
27         return modelo;
28     }
29     public void setModelo(String modelo) { no usages new *
30         this.modelo = modelo;
31     }
32
33     public float getPrecio() { no usages new *
34         return precio;
35     }
36
37     public void setPrecio(float precio) { no usages new *
38         this.precio = precio;
39     }
40
41     @Override new *
42     public String toString() {
43         return "SmartPhone{" +
44             "marca=" + marca + '\'' +
45             ", modelo=" + modelo + '\'' +
46             ", precio=" + precio +
47             '\'';
48     }
49 }
```


The screenshot shows the IntelliJ IDEA IDE with the project 'TareaX3_10_AppMVCVersion1' open. The project structure on the left includes a package 'mx.unam.fesa.ico' with sub-packages 'controller', 'model', and 'view'. The 'controller' package contains 'ControladorPrincipal.java', which is the active file. The code in the editor defines the 'ControladorPrincipal' class, which implements the 'MouseListener' interface. It includes imports for 'VentanaPrincipal', 'ModeloTablaPhone', 'SmartPhone', and standard Java event handling classes. The class has two private attributes: 'VentanaPrincipal view' and 'ModeloTablaPhone modelo'. The constructor 'ControladorPrincipal(VentanaPrincipal vista)' initializes these attributes and sets up mouse listeners for the 'view' object. The 'mouseClicked' method is implemented, which calls 'view.getBtnSaludar()' and 'view.getLblNombre()' to trigger actions in the view. The 'mousePressed' and 'mouseReleased' methods are also present but currently empty.

```
1 package mx.unam.fesa.ico.controller;
2
3 import mx.unam.fesa.ico.model.ModeloTablaPhone;
4 import mx.unam.fesa.ico.model.SmartPhone;
5 import mx.unam.fesa.ico.view.VentanaPrincipal;
6 import java.awt.event.MouseEvent;
7 import java.awt.event.MouseListener;
8 import java.util.ArrayList;
9 //mouse listener detecta el mouse y se implementan eventos del ratón
10
11 public class ControladorPrincipal implements MouseListener { 3 usages new *
12
13     private VentanaPrincipal view; 9 usages
14     private ModeloTablaPhone modelo; 2 usages
15
16     public ControladorPrincipal (VentanaPrincipal vista){ 1 usage new *
17         this.view = vista;
18         this.view.getBtnSaludar().addMouseListener( this);
19         this.view.getLblNombre().addMouseListener( this);
20         ArrayList<SmartPhone> tels = new ArrayList<>();
21         tels.add(new SmartPhone( marca: "Apple", modelo: "iPhone 15", precio: 17000.3f));
22         tels.add(new SmartPhone( marca: "Samsung", modelo: "Galaxy 5", precio: 7000.3f));
23         modelo = new ModeloTablaPhone(tels);
24         this.view.getTblPhone().setModel(modelo);
25         this.view.getTblPhone().updateUI();
26     }
27     //esta es una variable local al metodo, no el atributo ensi
28     @Override new *
```

This screenshot shows the same IntelliJ IDEA IDE, but now the 'mouseClicked' method in 'ControladorPrincipal.java' is being edited. The method signature is 'public void mouseClicked(MouseEvent e)'. The implementation calls 'this.view.getLblSalida().setText()' to update the output label with a greeting. It then checks if the source of the event is the 'getBtnSaludar()' button. If so, it prints 'Hola desde NezaYork' and 'Hola desde otra parte XD' to the console, and updates the 'getLblSalida()' label with 'Hola' followed by the text from 'getTxtNombre()'. Another check is made for the 'getLblNombre()' button, which prints 'Desde etiqueta 1' to the console. The 'mousePressed' and 'mouseReleased' methods are still empty.

```
11 public class ControladorPrincipal implements MouseListener { 3 usages new *
12     }
13     //esta es una variable local al metodo, no el atributo ensi
14     @Override new *
15     public void mouseClicked(MouseEvent e) {
16
17         //this.view.getLblSalida().setText("Hola " + this.view.getTxtNombre().getText());
18         if (e.getSource() == this.view.getBtnSaludar()){
19             System.out.println("Hola desde NezaYork");
20             System.out.println("Hola desde otra parte XD");
21             this.view.getLblSalida().setText("Hola " + this.view.getTxtNombre().getText());
22         }
23         if (e.getSource() == this.view.getLblNombre()){
24             System.out.println("Desde etiqueta 1");
25         }
26     }
27
28     @Override new *
29     public void mousePressed(MouseEvent e) {
30
31     }
32
33     @Override new *
34     public void mouseReleased(MouseEvent e) {
35
36     }
37
38     @Override new *
```

